

Reconstruction of the Profitability Factor in China

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ABSTRACT

We reconstruct the profitability factor in the Chinese sample using the Gross Profitability to Assets (GPA) ratio. Given the widespread presence of earnings management and policy distortions in Chinese financial statements, the profitability factor based on GPA is able to capture and mitigate these distortions. Compared to ROE, GPA provides a more comprehensive measure of profitability and more effectively captures the profitability effects in the Chinese market. Our findings show that both the gross profitability anomaly and the accrual anomaly are significantly present in China, exhibiting distinct characteristics before and after the split-share structure reform. The newly developed CH-5 factor model, constructed using GPA, successfully explains these two anomalies, captures the information contained in the earlier CH-3 model, and outperforms the traditional Fama-French five-factor (FF-5) model. Furthermore, during the anomaly explanation process, the GPA factor exhibits a “crowding-out effect” on the value factor, offering a novel framework for asset pricing research within the Chinese context.

1. Introduction

Asset pricing models are fundamental to modern financial theory and practice, reflecting the dynamic interplay between market anomalies and theoretical innovations. However, existing asset pricing models face significant challenges when applied to the Chinese market, particularly in constructing the profitability factor. The three-factor model proposed by [Fama and French \(1993\)](#), which consists of the market, size, and value factors, was the first systematic attempt to explain cross-sectional variations in stock returns. However, this model fails to fully account for market anomalies such as the profitability and investment effects ([Fama and French, 2015](#)). To address this issue, [Fama and French \(2015\)](#) extended the model by incorporating profitability and investment factors, leading to the five-factor model (FF-5), which quickly became a benchmark in asset pricing research. Recent evidence suggests that traditional models like FF-5 exhibit substantial instability across different segments of the A-share market, such as the STAR and second-board markets, primarily due to the noise inherent in accounting metrics ([Chen, 2024](#)).

Nevertheless, the construction of the profitability factor and its applicability to the Chinese market remain highly controversial, leaving a significant theoretical gap. In the FF-5 model, the profitability factor is typically measured using Return on Equity (ROE), which is calculated based on net income and shareholders' equity. However, in the Chinese market, due to policy interventions, earnings management, and differences in accounting standards, ROE fails to accurately reflect firms' true profitability. First, earnings management is prevalent among Chinese listed firms, where companies manipulate financial reports through related-party transactions, off-balance-sheet financing, and accrual adjustments ([Gao and Song, 2008](#)). Second, government subsidies are widely used in China as a tool for financial adjustments, particularly in non-state-owned enterprises, where they inflate reported profits and distort the stability of ROE ([Wang and Yung, 2011](#)). Additionally, discrepancies in accounting policies, such as asset valuation and revenue recognition, further weaken the comparability and reliability of ROE across firms and time periods. Consequently, the FF-5 model's explanatory power in the Chinese market is significantly impaired, leading to potential asset mispricing. Furthermore, the urgency of identifying authentic profitability is underscored by the China Securities Regulatory Commission's 2024 “Several Opinions of the State Council on Strengthening Supervision and Risk Prevention and Promoting the High-Quality Development of the Capital Market” (The New National Nine Articles) ([China Securities Regulatory Commission, 2024](#)). This landmark policy explicitly mandates a shift toward high-quality growth and sustainable dividend capacity. We argue that GPA, reflecting organic cash generation from top-line operations, serves as a far more reliable indicator of a firm's ability to meet these new regulatory standards and resist manipulation than bottom-line metrics.

The debate over the optimal measurement of profitability remains a central theme at the academic frontier. Even in developed markets, recent retrospectives underscore that the choice between top-line and bottom-line metrics is fundamental to capturing the true quality premium ([Novy-Marx and Medhat, 2025](#)). To overcome these limitations, researchers have explored alternative profitability measures. [Novy-Marx \(2013\)](#) proposed that gross profitability to asset ratio (GPA) better reflects firms' core profitability because it is less affected by non-cash expenses such as

depreciation and amortization, providing a more direct measure of firms' operating efficiency. [Ball et al. \(2016\)](#) further validated the advantages of gross profitability-based indicators, emphasizing their ability to deliver a "cleaner" profitability signal. However, despite its empirical success in the U.S. market, the applicability of the gross profitability factor in China remains underexplored. Existing factor models tailored for the Chinese market, such as CH-3 and Q-4, have not sufficiently examined profitability factors or their ability to explain market anomalies ([Liu et al., 2019](#); [Hou et al., 2015](#)).

Furthermore, prior research has identified notable profitability-related anomalies in the Chinese stock market, particularly the Gross Profitability Anomaly and the Accrual Anomaly ([Novy-Marx, 2013](#); [Sloan, 1996](#)). The Gross Profitability Anomaly suggests that firms with higher gross profitability tend to earn higher returns, even after controlling for size and value factors. Meanwhile, the Accrual Anomaly indicates that firms with higher accrual components in earnings typically experience lower future stock returns due to earnings manipulation. Both from an economic perspective and a practical standpoint, these excess return anomalies are closely related to financial statement factors in the Chinese market. Given the widespread financial reporting distortions in China, where government subsidies and accounting irregularities frequently distort firms' profitability signals, ROE-based profitability measures become unreliable.

Therefore, adopting a more robust profitability metric is crucial for improving factor pricing models in the Chinese context. This study aims to bridge this research gap by replacing ROE with GPA in the Fama-French five-factor model, enhancing its explanatory power and applicability to the Chinese market. Specifically, this study focuses on five key aspects: (1) Establishing the theoretical justification for using GPA as a superior profitability factor by integrating insights from asset pricing and accounting literature, followed by empirical validation using Fama-MacBeth regressions and the rigorous stepwise model selection framework recently advocated in top-tier literature ([Feng et al., 2026](#)). (2) Analyzing the unique characteristics of profitability effects in China, particularly from the perspectives of earnings management, financial statement manipulation, and government intervention. (3) Comparing the explanatory power of the GPA factor versus the traditional ROE factor, particularly in capturing profitability-related market anomalies. (4) Constructing an improved five-factor model tailored for the Chinese market and empirically verifying its effectiveness. Additionally, considering structural shifts in China's financial markets, such as the Stock Split Reform, this study examines pre- and post-reform periods to assess the robustness of the proposed model across different policy environments. (5) Finally, we conduct Long-Short regression tests on two market excess return anomalies in China and compare the ability of the CH-5 and FF-5 models in explaining abnormal returns.

This study contributes to the literature in four key ways: (1) Introducing and empirically validating GPA as a superior profitability factor for China, addressing a critical gap in the literature on profitability factor construction. (2) Developing the CH-5 model based on GPA and demonstrating its superior performance in explaining market anomalies compared to FF-5. (3) Providing robust evidence for the independence of the profitability factor from behavioral price signals; while the Chinese market is influenced by large individual trading behaviors creating a behavioral "Trend Factor" ([Liu et al., 2024](#)), our GPA factor captures a distinct fundamental risk premium. (4) Examining the impact of China's stock split reform on factor applicability, providing localized insights into asset pricing in emerging markets. By improving the accuracy of profitability measurement, this research enhances the effectiveness of factor models in China and offers valuable insights for both academic research and practical investment strategies.

2. Literature Review

Earnings management is a widespread phenomenon in the Chinese capital market, where firms manipulate financial data to enhance market valuation, meet regulatory requirements, or stabilize investor expectations ([Wang and Yung, 2011](#)). This practice is particularly evident in non-recurring income items such as asset disposal gains, government subsidies, and investment income. Firms exploit the non-recurring and manipulable nature of these items to influence net income-related metrics, such as return on equity (ROE), which plays a crucial role in investment analysis. Consequently, ROE can be significantly distorted, affecting investor assessments.

Earnings management primarily consists of accrual earnings management (AEM) and real earnings management (REM). AEM affects financial reporting through accounting policy choices and adjustments, such as depreciation methods and revenue recognition timing ([Healy and Wahlen, 1998](#)). In contrast, REM manipulates ROE by altering actual business operations, such as delaying expenditures or recognizing revenue prematurely ([Healy and Wahlen, 1998](#)). These practices often lead to short-term increases in ROE but may undermine a firm's long-term sustainability, reducing its reliability as an asset pricing factor ([Roychowdhury, 2006](#)). Research has shown that firms may artificially

boost ROE by cutting discretionary expenses such as research and development (R&D) and advertising costs (Bens et al., 2002, 2003; Dechow and Sloan, 1991).

Additionally, studies have identified related-party transactions (RPTs) as one of the most common tools for earnings management in China. Given the country's unique equity structure and corporate governance environment, controlling shareholders can manipulate financial data through revenue inflation, cost reduction, or asset transfers to meet profit targets or avoid delisting risks (Chen and Wang, 2005). Yu and Xia (2004) found that firms with concentrated ownership but weak governance mechanisms are more likely to engage in earnings management through related-party transactions, whereas firms with more dispersed ownership structures have stronger corporate governance and lower levels of earnings manipulation. Strengthening disclosure requirements for related-party transactions and enhancing the oversight role of independent directors are essential measures to mitigate the adverse effects of earnings management.

Previous studies have also found that government subsidies play a significant role in shaping the financial reporting quality of Chinese firms. Unlike developed markets, where subsidies are primarily allocated for industrial policy objectives, Chinese enterprises, particularly non-state-owned firms, often rely on government subsidies to improve their financial standing. Research has shown that subsidies can directly increase revenue and improve cash flow, making firms appear more profitable on financial statements and subsequently influencing investor perceptions (Bai et al., 2018). Bo and Wu (2009) further demonstrated that firms facing financial constraints or short-term performance pressure are more likely to engage in earnings management by leveraging government subsidies to inflate reported profits. Moreover, the combination of adjustments in accounting standards and government subsidies can further distort profitability metrics, thereby affecting the accurate assessment of firm performance. Xue et al. (2009) pointed out that changes in Chinese accounting standards have provided firms with greater flexibility for earnings management, allowing some companies to smooth profits through accounting rule modifications.

The study of profitability as an asset pricing factor can be traced back to Fama and French (2006), who found a significant positive relationship between profitability and stock returns, while asset growth exhibited a negative correlation with returns. By integrating the dividend discount model with net income accounting, they established the theoretical foundation for profitability as a core factor in explaining stock returns, demonstrating that firms with higher profitability generally yield higher expected returns.

Subsequently, Novy-Marx (2010) introduced gross profitability to assets (GPA) as an alternative measure of profitability, arguing that it serves as a superior predictor of stock returns compared to ROE. Unlike ROE, which can be significantly affected by earnings management and accounting policies, GPA directly measures a firm's operational efficiency. Furthermore, the recent retrospective by Novy-Marx and Medhat (2025) underscores that the debate over the optimal definition of profitability remains a central challenge at the academic frontier, reinforcing the necessity of re-evaluating metrics like GPA as global markets and accounting standards continue to evolve. Hou et al. (2015) further expanded the application of profitability factors by incorporating them into the q-factor model, reinforcing their importance in explaining stock return variations.

Fama and French (2015) validated the role of profitability factors by incorporating them into the five-factor model, significantly improving asset pricing accuracy by accounting for profitability and investment effects. Empirical research has confirmed that profitability remains a key determinant of stock returns across various markets (Feng et al., 2017). However, the applicability of profitability factors in the Chinese market remains an open question.

Empirical evidence suggests that the profitability effect in China exhibits unique characteristics. Gao and Zhou (2016) found that while profitability positively predicts stock returns among large firms, the effect is weaker for smaller firms, deviating from patterns observed in developed markets. One of the key challenges is that ROE, a commonly used profitability indicator in China, is highly susceptible to earnings manipulation. Firms frequently adjust depreciation schedules, alter revenue recognition methods, or rely on government subsidies to maintain high ROE figures, undermining its effectiveness as a stable profitability measure. Recent evidence by Chen (2024) specifically examines the robustness of the Fama-French model in China, finding significant instability across segments such as the STAR and second-board markets, which further motivates the search for more stable and fundamental pricing anchors like GPA.

To address these concerns, researchers have explored alternative profitability measures. Dickinson (2011) examined indicators such as return on net operating assets (RNOA), asset turnover (ATO), and profit margin (PM) to assess their effectiveness across different firm life cycle stages. Liu et al. (2019) identified that stock prices of small firms in China are influenced by "shell value," leading to the substitution of the earnings-to-price (EP) factor for the book-to-market (BM) ratio as a value factor, forming the CH-3 asset pricing model. However, their research did not specifically focus

on the profitability effect. Hou et al. (2021) introduced an expected growth factor to enhance factor model performance, while Hou et al. (2020) refined the q-factor model to better explain profitability-driven anomalies.

Recent research has further enriched the study of profitability effects. Gormsen and Lazarus (2019) showed that, after controlling for cash flow maturity, factor premia may not be solely driven by traditional factor characteristics such as profitability but are also influenced by the timing of cash flows, providing a new perspective on factor pricing. Detzel et al. (2023) emphasized the importance of the profitability factor, particularly cash profitability, in asset pricing models when accounting for transaction costs, demonstrating that it significantly improves the Sharpe ratio and enhances the model's ability to capture market anomalies. Additionally, Hu et al. (2025) introduced a novel factor perspective by examining investor behavior shifts due to uncertainty, particularly volatility uncertainty. They suggested that the profitability factor may be driven by firms with concentrated short-term cash flows, where investors place a higher premium on near-term cash flows, influencing the observed factor premium. Moreover, the independence of fundamental signals from behavioral patterns is a growing focus; Liu et al. (2024) identify a prominent "Trend Factor" in China driven by large individual trading, suggesting that isolating fundamental quality (like GPA) from speculative behavioral trends is crucial for robust model construction.

Given the unique characteristics of the Chinese market, this study argues that among the various alternatives to traditional profitability measures, GPA has emerged as one of the most promising solutions. Novy-Marx (2013) demonstrated that GPA outperforms traditional profitability metrics by mitigating distortions caused by financial leverage and accounting policies. While GPA has been widely validated in developed markets, its applicability in the Chinese market remains underexplored. Considering the prevalence of earnings management and strong government intervention in China, further research is needed to determine whether GPA can serve as a robust profitability measure within the local asset pricing framework.

3. Data Source and Sample

The empirical analysis in this study is based on a comprehensive dataset of Chinese A-share stocks listed on the Shanghai and Shenzhen Stock Exchanges. Our primary data source is the China Stock Market & Accounting Research (CSMAR) database, which provides high-quality financial statements, stock returns, and trading information. The sample period spans from January 2000 to November 2024, capturing diverse market cycles in China's evolving capital markets.

To ensure the reliability and consistency of the results, we applied a rigorous multi-stage data screening and preprocessing procedure. First, we excluded firms labeled as Special Treatment (ST) or those that have been delisted to avoid the impact of distressed financial conditions. Second, we removed observations with missing key financial variables (e.g., net income, total assets) or those with fewer than 15 trading days in a given month to ensure data completeness and liquidity. Third, to mitigate the "new-listing bias," firms listed for less than six months were excluded from the sample. Finally, we required each formed portfolio to contain at least 50 stocks to maintain statistical representativeness.

Following the initial screening, we addressed potential data issues through several standardized econometric techniques. To maintain a balanced panel, missing values were handled using linear interpolation where appropriate. To mitigate the influence of extreme outliers, all continuous variables were winsorized at the 1st and 99th percentiles; this threshold was determined after a manual inspection of the data distribution. Furthermore, before conducting the empirical regressions, we performed correlation tests and calculated Variance Inflation Factors (VIF) to detect potential multicollinearity, applying dimensionality reduction where necessary to ensure the parsimony of our model.

After these screening and cleaning procedures, our final dataset comprises approximately 620,000 firm-month observations. This robust sample provides a solid foundation for constructing our profitability factor and conducting the subsequent cross-sectional analyses.

4. Earnings Management and Government Subsidies

Earnings management is a prevalent phenomenon in the Chinese capital market, where firms manipulate financial data to achieve specific financial objectives, such as enhancing market valuation, meeting regulatory requirements, or stabilizing investor expectations (Wang and Yung, 2011). This is particularly common in non-operating income items such as asset disposal gains, government subsidies, and investment income. Due to the non-recurring and manipulable nature of these items, firms often resort to earnings management techniques to influence profit.

Earnings management encompasses two primary strategies: accrual-based earnings management (AEM) and real earnings management (REM). AEM involves manipulating financial reports through the selection and adjustment of accounting policies. This approach is particularly prevalent in China's capital markets, where companies often adjust non-operating items to inflate short-term profits, thereby enhancing reported earnings. Although these practices may comply with regulatory standards, they frequently fail to reflect the company's actual operational performance. Consequently, profitability indicators calculated based on net profit may be artificially inflated, undermining their effectiveness as asset pricing factors.

Beyond AEM, REM is another critical strategy, wherein managers alter actual business activities to adjust earnings levels. For instance, they might defer or accelerate expenditures, hasten revenue recognition, or even sell fixed assets to secure one-time gains, thereby boosting net profit. However, such actions typically lead to short-term profit increases at the potential expense of the company's long-term sustainability, rendering their impact temporary. Specifically, modifications in real business operations—such as adjusting working capital or delaying R&D investments—can distort profitability metrics, thereby affecting investors' accurate assessment of a company's earning capacity.

In the context of Chinese enterprises, related-party transactions represent one of the most common earnings management techniques. Within China's unique market environment and corporate governance framework, such practices are widespread and exhibit distinct characteristics, profoundly influencing firms' earnings quality and market valuation. Existing research has explored the patterns and economic consequences of these earnings management methods, particularly the differences between state-owned and non-state-owned enterprises. Additionally, studies have found that companies with higher controlling shareholder ownership are more prone to engage in earnings management, whereas firms with more dispersed ownership experience lower levels of earnings management due to the checks and balances inherent in their governance structures. To address this issue, enhancing disclosure requirements for related-party transactions and strengthening the supervisory role of independent directors are crucial measures to prevent listed companies from abusing such transactions for earnings management purposes.

Unlike other capital markets, government subsidies serve as a crucial fiscal policy instrument in China, holding a more prominent position in the country's capital market. The quality of financial reporting in China is significantly influenced by government subsidies, which not only create substantial differences in the treatment of non-recurring income items compared to other reporting practices but also provide opportunities for earnings management. This phenomenon is particularly evident among non-state-owned enterprises (non-SOEs), which frequently apply for and utilize government subsidies to manage earnings. These subsidies can be recognized as income during specific periods, thereby affecting current profitability. Consequently, the comparability of financial data across firms is reduced, ultimately impairing the stability of traditional profitability indicators.

Government subsidies influence financial statements by directly increasing revenue and improving cash flow, particularly affecting the income statement and balance sheet. These subsidies can distort short-term profitability indicators, making financial statements appear more favorable than they actually are. Non-SOEs, which face greater financing pressures and whose performance evaluations rely more heavily on short-term profits, are more inclined to engage in upward earnings management to enhance their accounting profit performance. Therefore, strengthening accounting oversight for non-SOEs, mitigating aggressive earnings management, and optimizing the performance evaluation mechanisms of state-owned enterprises (SOEs) to reduce incentives for earnings management are essential measures to improve the quality of financial reporting in Chinese enterprises. For instance, in the case of return on equity (ROE), the interaction effect between government subsidies and differences in accounting standards can be simplified as follows:

$$ROE = \frac{\pi_o + S + Q}{EQ} \quad (1)$$

where π_o is operating profit, S is government subsidies, Q represents net income changes due to accounting standard differences, and EQ is equity adjusted for differences in accounting standards. This interaction may lead to an overestimation of ROE, especially in cases where enterprises rely on government subsidies and calculate assets and income under different accounting standards, making the ROE error more significant.

Furthermore, the quality of financial statements is also influenced by accounting policy adjustments and managerial decisions. The implementation of new accounting standards in China may provide firms with greater flexibility in earnings management, leading some companies to exploit accounting standard changes to smooth earnings. Firms can also adjust profits through various methods, such as selecting different depreciation methods or altering the timing of

revenue recognition. While these adjustments comply with regulatory requirements, they may cause reported profits to deviate from the firm's actual operating performance, thereby undermining their reliability as profitability indicators. Consequently, financial statement quality issues, particularly in the treatment of non-recurring items and managerial discretion in accounting policy selection, have significant implications for the stability and effectiveness of return on equity.

The impact of earnings management can be better understood through real cases. In China's capital market, earnings management is a widespread phenomenon wherein companies manipulate financial data to achieve specific financial objectives. Metersbonwe Group (hereinafter referred to as "Metersbonwe"), a well-known company in China's apparel industry, faced declining business performance, with net profits turning negative at one point. In an effort to maintain its profitability and market image, Metersbonwe employed various earnings management strategies. First, the company relaxed its accounts receivable credit policy to increase sales revenue and improve its reported earnings. However, this practice heightened the risk of delayed receivables collection, adversely affecting the company's cash flow and long-term financial health. Second, the company generated one-time profits by transferring ownership of its subsidiaries, thereby boosting its current-period earnings. Although such non-recurring gains temporarily enhanced the financial statements, they did not reflect the company's long-term profitability. Additionally, Metersbonwe leveraged government subsidies as a profit-adjustment tool, recognizing them as part of its current earnings to inflate net profits. However, government subsidies are uncertain and unsustainable, and excessive reliance on them could weaken the company's market competitiveness. Although these earnings management techniques improved short-term financial indicators, they failed to reflect the firm's true operating condition, misled investors' decisions, and significantly harmed the company's long-term development, leading to severe stock price volatility and mispricing.

Within the income statement, profitability indicators positioned lower in the structure are more susceptible to distortion, as they incorporate more manipulable items than those higher up in the statement. This reduces their correlation with the firm's true economic profitability. As a result, net profit as a profitability metric is likely contaminated by earnings management and government subsidies in Chinese firms, raising questions about the appropriateness of using net profit-related indicators as a pricing anchor in asset pricing models for the Chinese market. Therefore, identifying a "cleaner" profitability factor that mitigates distortions from China-specific factors is of critical importance for current asset pricing research.

We do not intend to simply exclude firms with high levels of earnings management from our regression analysis to eliminate contamination effects, as these companies play a significant role in China's capital market. In the next section, we will develop our analysis based on the theoretical foundations of profitability effects, incorporating these potentially distorted firms into our regression analysis to further explore the applicability of profitability factors in the Chinese market.

5. Profitability in China

The research on profitability effects began with [Fama and French \(2006\)](#), who proposed a significant positive relationship between profitability and stock returns, while asset growth exhibited a negative correlation with stock returns ([Fama and French, 2006](#)). By integrating the dividend discount model with net income accounting, they demonstrated the logic that both the book-to-market ratio and profitability are positively related to expected returns. In the dividend discount model, the stock price equals the present value of its expected dividends. Under net income accounting, changes in book value are equal to retained earnings. The combination of these two implies that the market value of equity (including dividends) is represented by:

$$M_t = \sum_{\tau=0}^{\infty} \frac{E_t[Y_{t+\tau} - dB_{t+\tau}]}{(1+r)^\tau} \quad (2)$$

where Y_t is the return at time t , $dB_t = B_t - B_{t-1}$ is the change in book equity, and r is the required rate of return on expected dividends. All else being equal, a higher valuation implies a lower expected return, and a higher expected income implies a higher expected return. In other words, value stocks should outperform growth stocks, and profitable companies should outperform non-profitable companies.

This finding underscores the fundamental role of profitability and investment in predicting stock returns. Building on this foundation, [Fama and French \(2015\)](#) validated the role of the profitability factor through their five-factor model. By incorporating both the profitability and investment factors, the model further demonstrated the significance of

profitability in explaining variations in stock returns. Through extensive testing across diverse market samples, the introduction of the five-factor model significantly improved the accuracy of asset pricing models, particularly in capturing profitability differences across industries and firms.

Novy-Marx (2010) proposed that gross profitability to assets (GPA) serves as an effective proxy for profitability and can reliably predict stock returns. Unlike traditional profit measures based on net earnings, gross profitability better reflects a firm's long-term earning capacity, revealing that firms with higher profitability tend to generate significantly higher returns. His study suggests that gross profitability not only provides a clearer measure of a firm's financial health but also enhances the understanding of traditional value strategies. This effect is particularly pronounced in highly liquid markets, where the efficiency of profitability-based pricing mechanisms is more evident.

In the context of the Chinese market, previous studies have identified a unique pattern of the profitability effect: among small firms, those with higher profitability exhibit lower stock returns, whereas among large firms, highly profitable firms tend to achieve higher stock returns. Compared to developed capital markets such as the United States, the applicability of the profitability factor in China is influenced by market structure and firms' financial characteristics. Based on the theoretical inferences above, we hypothesize that the prevalence of earnings management among small firms contributes to the weakened pricing efficiency in this segment of the market.

Although Fama and French (2006) and subsequent studies have confirmed the effectiveness of return on equity as a profitability factor in international markets, its long-standing role as a core metric for measuring profitability has also made it widely used in asset pricing research in China. However, given the widespread presence of earnings management among Chinese firms, ROE may be susceptible to financial manipulation, preventing it from accurately reflecting firms' true profitability. For instance, companies may optimize their ROE by adjusting depreciation schedules, manipulating revenue recognition practices, or relying on government subsidies, thereby compromising its stability as a profitability factor.

By contrast, gross profitability is considered a more robust measure of firm profitability. Novy-Marx (2013) found that the GPA factor more effectively captures firms' fundamental profitability and exhibits strong predictive power for stock returns in the U.S. market. As gross profitability is positioned at the top of the income statement, it is directly derived from revenue and cost of goods sold, thereby avoiding distortions introduced by financial leverage, depreciation policies, and other discretionary accounting choices. Consequently, it provides a more reliable measure of firms' long-term profitability.

Although GPA has been validated in international markets, there remains a lack of systematic empirical analysis in the Chinese market. Given the distinct accounting environment and earnings management practices in China, it is essential to investigate whether GPA can play a similar role in the local asset pricing framework. This study aims to fill this research gap by further examining the applicability and robustness of the profitability effect in the Chinese market.

In this chapter, we compare Fama-MacBeth regressions and perform a "horse racing" analysis of the underlying variables in the factor construction. The Fama-MacBeth regression is a classic cross-sectional regression method that effectively eliminates time-series autocorrelation and heteroscedasticity issues, making it suitable for cross-sectional data analysis. The regression analysis includes commonly used valuation ratios in the Fama-French five-factor model construction and introduces GPA as an alternative profitability factor. We adopt the same methodology as the previous study and conduct a Fama-MacBeth (1973) cross-sectional regression on the monthly excess returns of individual stocks (Fama and MacBeth, 1973). The regression factors include the estimated CAPM beta coefficient (β) of the stock, the size factor, and the profitability factor.

Table 1 reports the average slope coefficients and t-values from the monthly Fama-MacBeth regressions when each valuation ratio is individually included in the regression model. Consistent with prior findings for both the Chinese and U.S. markets, the coefficient on the market factor (Beta) is close to zero and statistically insignificant across most specifications. This reflects a well-documented feature of Fama-MacBeth cross-sectional regressions: when the cross-sectional variation in returns is weakly related to market risk, the estimated Beta naturally tends to be small, even if the market factor is important in a time-series setting (Fama and French, 1993; Liu et al., 2019). Consistent with findings from the U.S. market, the results in column (1) indicate that the Beta coefficient does not significantly impact stock returns. However, when logME (firm size) and BM (book-to-market ratio) are introduced in the regression model, the Beta coefficient becomes significantly negative, while the size variable exhibits strong explanatory power, with a t-value of 4.85. These findings further confirm the existence of a significant size effect in the Chinese market.

In columns (4) and (5), we introduce the profitability factor and the investment factor into the regression. The average slope of ROE is negligible, and it is accompanied by a low-confidence t-value. While the investment factor is statistically significant, its average slope coefficient is negative, aligning with the findings of previous studies.

Table 1

Fama-MacBeth regressions of stock returns on valuation factors.

The table reports average slope coefficients from month-by-month Fama-MacBeth regressions. Individual stock returns are regressed cross-sectionally on stock characteristics as of the previous month. The first to fifth columns present the regression results based on the factors used in the Fama-French five-factor (FF-5) model. The sixth and seventh columns report the regressions employing the CH-5 factors. The eighth column displays the results of a horse race regression that incorporates all factors together. The columns correspond to different regression specifications, with nonempty rows indicating the included regressors. The sample period is January 2000 through December 2024. Our sample data includes only stocks codes starting with 600, 000, or 300.

Quantity	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Intercept</i>	0.002483 (1.16)	-0.074956 (-4.91)	-0.070916 (-4.63)	-0.071146 (-4.66)	-0.072902 (-4.84)	-0.068589 (-4.55)	-0.070397 (-4.73)	-0.070520 (-4.75)
<i>Beta</i>	0.004875 (1.96)	-0.035847 (-2.80)	-0.033692 (-2.65)	-0.033865 (-2.68)	-0.035598 (-2.86)	-0.032534 (-2.63)	-0.034332 (-2.82)	-0.034442 (-2.83)
<i>logME</i>	- (4.85)	0.006557 (4.65)	0.006244 (4.68)	0.006258 (4.92)	0.006462 (4.53)	0.005895 (4.77)	0.006109 (4.78)	0.006114 (4.78)
<i>BM</i>	- (-3.88)	- (-3.88)	-0.005025 (-3.88)	-0.005025 (-3.72)	-0.004922 (-3.82)	-0.004777 (-3.62)	-0.004638 (-3.62)	-0.004637 (-3.62)
<i>ROE</i>	- (-0.61)	- (-0.61)	- (-0.61)	-0.000248 (-0.26)	-0.000100 (-0.26)	- (-0.26)	- (-0.26)	-0.000164 (-0.47)
<i>GPA</i>	- (3.54)	- (3.54)	- (3.54)	- (3.54)	- (3.54)	0.060941 (3.54)	0.062070 (3.63)	0.062040 (3.64)
<i>INV</i>	- (-2.22)	- (-2.22)	- (-2.22)	- (-2.22)	-0.003434 (-2.22)	- (-2.22)	-0.003790 (-2.52)	-0.003772 (-2.52)

Columns (6) and (7) present the regression results when GPA replaces ROE as the profitability factor. Compared to ROE, GPA demonstrates superior explanatory power. Particularly in column (7), where GPA serves as the primary competitor, its average slope reaches 0.062070, making it one of the most dominant factors among the five explanatory variables. The t-statistic of GPA is 3.63, indicating that GPA is a more effective profitability factor in the Chinese market sample, capable of explaining a substantial portion of individual stock excess returns.

When all six valuation ratios are included in the regression model (column (8)), GPA remains the most dominant factor. In contrast, ROE exhibits an average slope of only -0.000164 with a t-statistic of -0.47, demonstrating that regardless of whether it is included in a “horse race” against GPA, ROE fails to explain stock returns, whereas GPA consistently exhibits superior explanatory power compared to other factors. These results remain robust even after excluding ChiNext and Shenzhen market stocks, as evidenced by the robustness tests presented in the Appendix.

Overall, our findings suggest that GPA is a more stable measure for capturing market return variations and exhibits stronger explanatory power in excess return prediction compared to ROE. Consequently, in the next section, we construct a five-factor model tailored for the Chinese market, replacing ROE with GPA as the profitability factor. The superiority of GPA over ROE is further substantiated in this part, where we present descriptive statistics and redundancy tests. Our results demonstrate that the CH-5 model, based on GPA, can successfully price assets, whereas the traditional FF-5 model, based on ROE, fails to do so.

6. CH-5 factors model

6.1. Construction of CH-5

Fama and French (2015) proposed the following five-factor model:

$$R_{it} - R_{Ft} = \alpha_i + \beta_i(R_{Mt} - R_{Ft}) + s_iSMB_t + h_iHML_t + r_iRMW_t + c_iCMA_t + \epsilon_{it} \quad (3)$$

where R_{it} is the return of portfolio i in period t , R_{Ft} is the risk-free rate, R_{Mt} is the market portfolio return, which is the value-weighted average return of the market, and $R_{Mt} - R_{Ft}$ reflects the market risk premium. SMB_t is the size factor, which represents the return difference between small-cap and large-cap stocks. HML_t is the value factor, which represents the return difference between high book-to-market and low book-to-market stocks. RMW_t is the profitability factor, representing the return difference between stocks with high operating profit margins and those with low operating profit margins. CMA_t is the investment factor, representing the return difference between stocks with conservative and aggressive investment styles.

In a manner similar to Fama and French (2015), this study groups stocks based on the quantiles of each indicator at the end of June in year t , to determine the stock portfolios for the period from July in year t to June in year $t + 1$. For stock i , its market capitalization at the end of June in year t is used as the $Size_{it}$ indicator; the “book-to-market ratio” ($BM_{i,t}$) is calculated by dividing the book value at the end of year $t - 1$ by the market capitalization of stock i at the end of December in year $t - 1$. Given the differences in market environments between China and the U.S., this study uses the “Gross Profitability to Total Assets” ratio to replace the “Operating Profit Margin” from the original model. The “Investment Style” (Inv_{it}) is calculated by the change in total assets from the end of year $t - 2$ to the end of year $t - 1$, divided by the total assets at the end of year $t - 2$.

To ensure the reliability of our pricing signals and avoid contamination from “shell value” or liquidity issues, we follow Liu et al. (2019) by excluding the bottom 30% of small-cap stocks before the grouping process. The risk factors are then constructed using the classic 2×3 sorting approach. Specifically, we first divide the remaining stocks into small-cap (S) and large-cap (B) groups based on the median market capitalization. Independently, stocks are sorted into three groups based on the 30th and 70th percentiles of their book-to-market ratio (High, Neutral, Low), our reconstructed GPA-based profitability (Robust, Neutral, Weak), and investment growth (Conservative, Neutral, Aggressive).

The final factors are derived from the returns of these intersecting portfolios. The size factor (SMB) is calculated as the average return difference between small-cap and large-cap portfolios across the three characteristic dimensions (BM, GPA, and Investment), thereby neutralizing the influence of other factors on the size premium. Similarly, the value (HML), profitability (RMW), and investment (CMA) factors are computed as the average return differences between the high-ranking and low-ranking characteristic portfolios within both the small-cap and large-cap segments. This independent sorting and averaging procedure ensures that each factor captures its respective risk premium while controlling for the influence of other variables.

Table 2 presents the descriptive statistics of the five factors. All data are reported in monthly percentage terms, covering the sample period from January 2000 to December 2024, spanning a total of 299 months. A detailed statistical analysis is performed on the market factor (MKT), size factor (SMB), book-to-market factor (HML), profitability factor (RMW), and investment factor (CMA).

The market factor (MKT) has an average monthly return of 0.671%, indicating that the overall market return was positive during the sample period, reflecting positive market performance. However, the standard deviation of the market factor is 7.588%, indicating considerable volatility in market returns and higher risk. The t-statistic is 1.53, which is close to the critical value of 1.96, suggesting that the return of the market factor is statistically significant, but with weaker significance.

The size factor (SMB) has an average monthly return of 0.457%, showing that small-cap stocks performed relatively better during the sample period. The standard deviation is 3.926%, reflecting substantial volatility in the returns of small-cap stocks, indicating higher risk. The t-statistic is 2.01, suggesting strong statistical significance, confirming the presence of a significant size effect during the sample period.

The book-to-market factor (HML) has an average monthly return of 0.311%, indicating that, on average, value stocks underperformed during the sample period. The standard deviation is 3.858%, showing considerable volatility in the returns of value stocks, with higher associated risk. The t-statistic is 1.39, suggesting that the return of the value

Table 2

Summary statistics for the CH-5 factors.

This table reports the means, standard deviations, t-statistics, and pairwise correlations for the three factors in the CH-5 model. The means and standard deviations are expressed in percent per month. The sample period is January 2000 through November 2024 (299 months).

Descriptive Statistics and Correlation Coefficient Matrix								
Factor	Mean	Std. dev.	t-stat	Correlations				
				MKT	SMB	HML	RMW	CMA
MKT	0.671	7.588	1.53	1.000	0.183	-0.098	-0.322	-0.007
SMB	0.457	3.926	2.01		1.000	-0.374	-0.319	0.198
HML	0.311	3.858	1.39			1.000	-0.437	0.500
RMW	0.277	3.084	1.56				1.000	-0.578
CMA	0.048	2.228	0.37					1.000

factor is statistically close to significance, but not fully significant. This implies that the return effect of the value factor was moderate during the sample period.

The profitability factor (RMW) has an average monthly return of 0.277%, indicating that companies with higher profitability generated positive returns during the sample period. The standard deviation is 3.084%, showing moderate volatility in the returns of the profitability factor. The t-statistic is 1.56, close to the critical value of 1.96, indicating that the return of this factor is statistically significant. The profitability factor exhibited relatively stronger return effects compared to other factors, further emphasizing the importance of profitability in explaining stock returns.

The investment factor (CMA) has an average monthly return of 0.048%, indicating that the return of the investment factor was relatively low during the sample period. The standard deviation is 2.228%, showing lower volatility and risk compared to other factors. The t-statistic is 0.37, indicating that the return of this factor is statistically insignificant. This suggests that the investment factor's return effect was weak during the sample period and failed to exhibit a significant investment style premium, consistent with the findings of prior research.

6.2. “Split-share structure” policy in China

Table 3 presents the mutual spanning matrix of the expanded six-factor model. Although the primary focus of this study is the construction of the CH-5 model based on the GPA profitability factor, we incorporate the momentum factor (MOM) into the spanning test to ensure the robustness of our framework. Given that momentum is a widely documented anomaly in global equity markets, it is crucial to examine whether our proposed factors, particularly the newly reconstructed RMW, provide independent pricing information that is not subsumed by price trends.

The results for the period from 2000 to 2024 indicate that, after controlling for other risk factors, significant market risk, size effect, book-to-market effect, and profitability effect remain in the A-share market, with t -statistics of 2.80, 5.44, 5.26, and 5.25 respectively. Notably, our GPA-based profitability factor (RMW) maintains a highly significant alpha (0.620%, $t = 5.25$) even when the momentum factor (MOM) is included in the spanning set. However, the regression intercept for the CMA factor cannot reject the null hypothesis that the intercept is zero ($t = -0.02$), implying that the CMA factor is “redundant” within this framework. Similarly, the MOM factor yields an insignificant alpha ($t = 0.97$). This finding is consistent with the results of scholars such as Li et al. (2017).

Due to the uniqueness of the Chinese sample, we hypothesize that the redundancy test results may be significantly influenced by the early data. The A-share market had the unique issue of “split share structure” in its early years, an inherent institutional flaw that distorted the pricing mechanism (Wu, 2006; Liu et al., 2010). The split share structure reform officially started on April 29, 2005, and by the end of December 2006, the reform task was essentially completed. Therefore, we split the sample into two sub-periods for further analysis.

Table 4 presents the results before and after the reform. The market risk premium was highly significant before the reform but became insignificant after. In contrast, the significance of the size and profitability effects significantly

Table 3

Mutual Spanning Matrix of the CH-5 and Momentum Factors. This table reports the results of mutual spanning regressions where each factor is regressed against the remaining factors in the six-factor framework (MKT, SMB, HML, RMW, CMA, and MOM). The constant represents the estimated monthly alpha (%). Heteroskedasticity-consistent *t*-statistics are reported in parentheses. The sample period is January 2000 through November 2024 (299 months).

Factor	MKT	SMB	HML	RMW	CMA	MOM
const	1.334 (2.80)	0.860 (5.44)	0.721 (5.26)	0.620 (5.25)	-0.002 (-0.02)	0.419 (0.97)
MKT	– (–)	-0.024 (-0.80)	-0.070 (-2.41)	-0.120 (-6.67)	-0.036 (-1.89)	-0.027 (-0.33)
SMB	-0.141 (-0.82)	– (–)	-0.533 (-9.87)	-0.296 (-7.40)	0.145 (4.14)	-0.389 (-2.01)
HML	-0.546 (-2.65)	-0.710 (-10.00)	– (–)	-0.391 (-8.15)	0.264 (6.14)	0.318 (1.49)
RMW	-1.402 (-5.56)	-0.591 (-6.22)	-0.588 (-6.39)	– (–)	-0.223 (-4.21)	0.343 (1.51)
CMA	-0.643 (-1.97)	0.443 (3.96)	0.608 (5.74)	-0.341 (-4.61)	– (–)	-0.648 (-1.95)
MOM	-0.024 (-0.32)	-0.060 (-2.14)	0.037 (1.48)	0.026 (1.53)	-0.033 (-2.06)	– (–)
adj. R^2	0.188	0.482	0.597	0.580	0.475	0.143

Note: *t*-statistics are reported in parentheses.

Table 4

Constants of the CH-5 Factors Redundancy Test in different periods.

Period	MKT	SMB	HML	RMW	CMA
2000.01 - 2007.06	2.044 (2.477)	0.407 (1.905)	0.842 (3.340)	0.358 (2.326)	0.014 (0.110)
2007.07 - 2024.11	0.863 (1.648)	0.923 (4.651)	0.695 (4.177)	0.674 (4.719)	-0.073 (-0.611)

improved after the stock reform. The intercept of the RMW factor increased from 0.358 to 0.674, indicating that the portion of these effects that could not be explained increased significantly, making the profitability factor more similar to those in developed markets.

6.3. The 5x5 portfolios return

To investigate the impact of profitability on returns, we construct three 5×5 factor portfolios based on the GPA profitability metric, calculating the weighted average returns for these portfolios. These portfolios are constructed based on the quantiles of three factors: market equity (ME), book-to-market ratio (BM), and investment (INV). Each factor is divided into five groups, and within these five groups, stocks are further sorted based on GPA in ascending order, classified as Weak, 2, 3, 4, and Robust. The results are presented in [Table 5](#), where we observe a clear upward trend in returns from the Weak to the Robust group.

Specifically, in the ME-GPA portfolio, the value-weighted return shows a clear upward trend as GPA increases. This trend suggests that portfolios with stronger profitability (higher GPA) outperform those with weaker profitability, with the average return difference between the Robust and Weak groups being 0.4155. The maximum difference is observed in the second row, indicating that the profitability effect of GPA is most pronounced in this group. Additionally, similar upward trends are observed in the portfolios based on BM and INV. The difference in returns for the BM-GPA portfolio

Table 5

The chart displays the weighted average returns for the 25-portfolio groups. For each factor (ME, BM, INV), we divide the sample stocks into five groups based on the quantiles of the factor. The groups labeled as Weak to Robust are based on the GPA (Gross Profitability to Asset) factor, sorted in ascending order. The weighted average returns for each group are then calculated using market capitalization. Individual stock returns are calculated in percentage terms. Our sample period is January 2000 through November 2024.

MEgroup5	Weak	2	3	4	Robust	Robust - Weak
Small	1.1443	1.3066	1.2523	1.5027	1.5514	0.4071
2	0.7592	1.0871	1.1470	1.0771	1.3502	0.5910
3	0.7131	0.8430	0.8773	1.0385	1.0420	0.3289
4	0.5821	0.7010	0.8170	1.0060	1.0010	0.4189
Big	0.4593	0.4213	0.6040	0.4640	0.7910	0.3317
BMgroup5	Weak	2	3	4	Robust	Robust - Weak
Low	0.0029	0.1035	0.5197	0.6260	0.7201	0.7172
2	0.6855	0.4175	0.3237	0.7602	1.0750	0.3895
3	0.7383	0.6683	0.8960	0.7514	1.0898	0.3515
4	0.6104	1.0464	0.9332	1.0495	0.7553	0.1449
High	0.8250	0.7974	1.0401	0.9243	1.0310	0.2060
INVgroup5	Weak	2	3	4	Robust	Robust - Weak
Conservative	0.7739	0.7165	0.6760	0.8844	0.9170	0.1431
2	0.4301	0.7090	0.7784	0.8420	1.0033	0.5732
3	0.7459	0.8911	0.8837	0.6980	0.9071	0.1612
4	0.4820	0.4514	0.7810	0.6711	1.1873	0.7053
Aggressive	0.4290	0.5821	0.6340	0.7820	0.4562	0.0272

has an average value of 0.3618, but it does not exhibit the same linear relationship as the ME-GPA group. Instead, it demonstrates an increasing volatility pattern. The performance of the INV-GPA portfolio shows a similar trend, though with slightly lower overall volatility compared to the BM-GPA portfolio, and an average difference of 0.3220.

This inconsistency suggests that while GPA serves as an effective profitability indicator in most cases, other factors such as market capitalization, book-to-market ratio, and investment style may significantly influence returns in certain circumstances. Therefore, when conducting factor analysis, it is essential to not only consider GPA but also take into account other variables, such as market environment and firm characteristics, to provide a more comprehensive evaluation of the determinants of stock returns.

Although profitability effects show volatility across different grouping methods, a clear profitability effect is evident across all groupings. The data clearly show that as GPA increases from Weak to Robust, returns rise significantly, validating the effectiveness of GPA as a profitability indicator. This further supports the rationale for using GPA as a profitability assessment tool in the Chinese market. Additionally, this table also confirms the characteristics of size, value, and investment effects in the Chinese sample, with results consistent with previous studies. These results are not discussed further in this paper.

In summary, this section empirically demonstrates the effectiveness of GPA as a profitability indicator across different policy periods, particularly in the upward trend of returns. The next section of this paper will focus on comparing the CH-5 model and the FF-5 model, specifically highlighting the advantages of the GPA factor.

6.4. Explaining Each Other's Factors

Table 6 presents the ability of the CH-5 and FF-5 models to explain each other's factors, alongside an extended spanning test incorporating momentum. In Panel A, we perform regression analyses to explain the market capitalization and profitability factors of the opposing model. The monthly alpha of the CH-5 model relative to the FF-5 profitability

Table 6

Abilities of different models to explain factor returns and Factor Spanning Tests.

Panel A reports a factor's estimated monthly alpha (in percentage) with respect to different models, with [White \(1980\)](#) heteroskedasticity-consistent t -statistics in parentheses. The models include the baseline CH-5, the Fama–French five-factor (FF-5) model, and an extended CH-5 model incorporating the momentum factor (CH-5 + MOM). Panel B computes the Gibbons–Ross–Shanken (1989) F -test of whether a given model produces zero alphas for the factors of the other model (p -value in parentheses). The sample period is January 2000 through November 2024 (299 months).

Panel A: Alpha (t-statistic)			
Alphas with respect to:	CH-5	FF-5	CH-5 + MOM
FFSMB	0.008 (0.66)	– –	– –
FFRMW	0.045 (0.50)	– –	– –
SMB	– –	0.019 (2.54)	– –
RMW	0.637 (5.48)	0.359 (3.89)	0.620 (5.25)
Panel B: GRS F-statistics (p-value)			
Alphas with respect to:	CH-5	FF-5	–
FF(SMB, HML, RMW, CMA)	0.21 (0.93)	– –	– –
SMB, HML, RMW, CMA	– –	3.77 (0.005)	– –

factor is 0.045%, with a t -statistic of 0.50, indicating that the CH-5 model effectively captures the information contained in the FF-5 profitability factor.

In contrast, the monthly alpha of the FF-5 model with respect to our GPA-based RMW factor is 0.359% ($t = 3.89$), which is statistically significant at the 1% level. This suggests that the FF-5 model fails to capture the unique risk premium associated with our reconstructed profitability factor, thereby highlighting its independent pricing power. The market capitalization factor exhibits similar characteristics; the unexplained component under the FF-5 model is approximately twice as large as that under the CH-5 model, a result primarily driven by the incremental information provided by the GPA-based RMW factor.

To further ensure that the pricing power of our profitability factor is not a proxy for price momentum, we perform an extended factor spanning test in the third column of Panel A. Even after incorporating the momentum factor (MOM) into the baseline CH-5 model, the monthly alpha for the RMW factor remains highly significant at 0.620% with a t -statistic of 5.25. This robust result provides empirical evidence that the profitability premium captured by our model is an independent phenomenon and is not subsumed by the momentum effect.

In Panel B, we conduct the Gibbons–Ross–Shanken (GRS) F -test to assess whether the alphas of a model's factors are jointly zero under the alternative model ([Gibbons et al., 1989](#)). The results further support our conclusions: the p -value for the joint alpha of the CH-5 factors when applied to the FF-5 model is 0.93, suggesting that the CH-5 model encompasses the factors of the FF-5 model well. In contrast, the p -value for the joint alpha of the FF-5 factors under the CH-5 model is 0.005, significantly rejecting the null hypothesis and indicating that the FF-5 model fails to capture the factor structure of the CH-5 model, particularly the RMW factor.

Beyond the comparison with the FF-5 model, we also extend this analysis to the CH-3 model of [Liu et al. \(2019\)](#). The results show that the CH-5 model effectively captures the information contained in CH-3, as indicated by small and insignificant alphas. In contrast, the CH-3 model fails to fully explain the CH-5 factors, with significant alphas remaining for SMB and RMW ($t = 2.60$ and $t = 2.69$, respectively). The GRS test confirms that CH-5 adequately explains CH-3 ($p = 0.18$), while CH-3 is rejected against CH-5 ($p = 0.0002$). For transparency, detailed results are reported in Appendix A2, [Table 12](#).

In practice, these experimental results clearly demonstrate that the RMW factor constructed using GPA in the CH-5 model possesses significant and independent pricing power. It outperforms both the traditional Fama–French five-factor model and the earlier CH-3 framework. For investors and researchers, the GRS and spanning tests serve as crucial tools for model selection, confirming that the CH-5 model provides a more scientific and robust basis for asset pricing and risk management in the Chinese context.

7. Anomaly in China

A standard for evaluating a factor model lies not only in its ability to price the factors of another model but also in the extent to which it refines the existing framework. In this section, we highlight the optimization introduced by the CH-5 model relative to the FF-5 model. Rather than attempting to capture all possible anomalies, the CH-5 model specifically improves the profitability dimension by replacing the traditional RMW factor with a measure designed to mitigate the influence of earnings management in Chinese financial statements.

Given the unique features of the Chinese market, we focus on two profitability-related anomalies: the Gross Profitability Anomaly and the Accrual Anomaly. We first confirm the presence of these anomalies using CAPM long–short portfolios and then examine how the alphas change when moving from CAPM to the CH-5 and FF-5 models. The results show that CH-5 consistently delivers lower and more stable alphas for these anomalies compared to FF-5. This demonstrates that CH-5 provides a more refined structure for the Chinese setting. Importantly, the improvement does not arise from arbitrarily expanding the factor space but rather from a targeted optimization of the profitability factor, which better aligns with the economic characteristics of the local market.

7.1. Gross profitability anomaly

[Novy-Marx \(2013\)](#) first introduced the gross profitability anomaly, which states that companies with high gross profitability (calculated as operating income minus operating costs, divided by total assets) tend to have significantly higher average returns than companies with low gross profitability. This finding challenges the value premium explanation proposed by [Fama and French \(1993\)](#), which argues that high returns come from low-profitability value stocks. Novy-Marx, however, pointed out that companies with high gross profitability are typically associated with sustained growth in profits, cash flows, and dividends. Despite having longer cash flow durations and lower operating leverage, these companies still achieve higher expected returns.

[Table 7](#) presents the results of the gross profitability anomaly test over various time periods, comparing the explanatory power of the CAPM, CH-5, and FF-5 asset pricing models under market capitalization-weighted portfolios as well as market capitalization-weighted portfolios excluding the bottom 30% of small-cap stocks. First, the CAPM model shows significant alpha values in all panels, with coefficients of 1.30 and 1.12, respectively, validating the significant existence of the gross profitability anomaly in the Chinese market. This indicates that stocks with high gross profitability continue to generate abnormal returns even after risk adjustment.

The RMW factor in both models demonstrates significant superiority in explaining the gross profitability anomaly. In Panel B, the RMW factor coefficients in the CH-5 and FF-5 models are 1.34 and 0.92, respectively, which are much higher than the portions explained by other factors, with t-values of 17.08 and 8.37. When regressing without other factors, the t-values of RMW are 27.09 and 17.04, showing the outstanding performance of the RMW factor in capturing firm profitability. This advantage remains robust in the weighted portfolios after excluding small-cap stocks.

Additionally, the difference between the CH-5 and FF-5 models in explaining the gross profitability anomaly is worth noting. While both models can effectively explain the gross profitability anomaly, the RMW factor coefficient and t-value in the CH-5 model are higher than those in the FF-5 model, indicating that the CH-5 model may better capture the profitability premium in the Chinese market. Upon examining the results, we find that this difference is mainly accounted for by the HML factor. The RMW factor, constructed using GPA, captures part of the excess returns that could have been explained by the HML factor, rendering the HML factor insignificant. This suggests that the GPA factor exhibits both value and profitability characteristics.

Table 7

Alphas and factor loadings for gross profitability anomaly.

The table reports the monthly long-short return spread's alpha and factor loadings. The regression estimated is:

$$R_t = \alpha + \beta_{MKT} MKT_t + \beta_{SMB} SMB_t + \beta_{HML} HML_t + \beta_{RMW} RMW_t + \beta_{CMA} CMA_t + \epsilon_t$$

where R_t is the anomaly's long-short return spread in month t , MKT_t is the excess market return, SMB_t is size factor, and HML_t is the BM-based value factor, CMA_t is based on the investment factor using total assets, RMW_t is constructed using both ROE and GPA, but in this table, RMW_t is used to represent both. In Panel A, we use the market cap weighted method for sorting. In this approach, the long leg of the Gross profitability anomaly is the value-weighted portfolio of stocks in the highest decile of the anomaly measure, and the short leg contains the stocks in the lowest decile. A high value of the anomaly is associated with higher returns, so the long-short portfolios create a return spread. We use regression to examine the statistical significance of this return spread. In Panel B, we first remove the bottom 30% of the smallest stocks to avoid contamination from shell value and other small-cap effects. After this, we proceed with market cap weighting on the remaining sample. Our sample period is January 2000 through November 2024. All t-statistics are based on the heteroskedasticity-consistent standard errors of [White \(1980\)](#).

Model	α	β_{MKT}	β_{SMB}	β_{HML}	β_{RMW}	β_{CMA}	$t(\alpha)$	$t(\beta_{MKT})$	$t(\beta_{SMB})$	$t(\beta_{HML})$	$t(\beta_{RMW})$	$t(\beta_{CMA})$
Panel A: Market Cap Weighted												
CAPM	1.30	-0.25	–	–	–	–	3.81	-5.50	–	–	–	–
CH5	0.58	-0.01	–	–	1.63	–	3.13	-0.40	–	–	25.78	–
	0.87	-0.01	-0.42	-0.04	1.36	-0.17	5.01	-0.54	-7.36	-0.65	16.77	-1.77
FF5	0.89	-0.04	–	–	1.39	–	3.83	-1.36	–	–	17.89	–
	1.40	-0.09	-0.48	-0.55	0.93	-0.03	6.50	-2.99	-5.88	-7.49	8.28	-0.23
Panel B: Market Cap Weighted After Removing 30% Smallest Size												
CAPM	1.12	-0.27	–	–	–	–	3.39	-6.23	–	–	–	–
CH5	0.42	-0.04	–	–	1.59	–	2.39	-1.60	–	–	27.09	–
	0.70	-0.05	-0.37	-0.10	1.34	-0.10	4.18	-1.96	-6.79	-1.57	17.08	-1.05
FF5	0.73	-0.08	–	–	1.31	–	3.18	-2.44	–	–	17.04	–
	1.23	-0.12	-0.44	-0.61	0.92	0.04	5.83	-4.13	-5.46	-8.41	8.37	0.33

This “crowding-out” effect implies that the information traditionally captured by the HML factor is now being absorbed by the GPA-based RMW factor. We interpret this phenomenon from both accounting and asset pricing perspectives.

First, gross profitability inherently contains valuation-related signals. Firms with high GPA are often undervalued relative to their earnings power, trading at modest book-to-market ratios but delivering strong future returns. This supports Novy-Marx’s (2013) view that gross profitability can outperform or complement traditional value indicators.

Second, in the Chinese market, the book-to-market ratio suffers from several structural distortions. Government subsidies, differences in accounting standards, and widespread earnings management reduce the reliability of book value as a measure of intrinsic firm worth. GPA, calculated from top-line items such as revenue and cost of goods sold, avoids many of these issues and offers a “cleaner” signal of firm quality. As a result, GPA not only captures profitability but also unintentionally substitutes for value-related information, thereby weakening the statistical significance of the HML factor when both are included in the model.

Therefore, the crowding-out effect arises not from multicollinearity but from a fundamental overlap in the economic roles of GPA and HML. In emerging markets with noisy accounting data, cleaner profitability measures like GPA can dominate traditional valuation ratios in explaining cross-sectional stock returns.

Drawing on the perspective of [Novy-Marx \(2013\)](#), from an economic standpoint, companies with high gross profitability not only generate higher returns but also have stock prices that better reflect their intrinsic value, avoiding the low-quality asset dilemma often faced by traditional value strategies. Thus, combining GPA with value effects not only allows investors to enjoy higher returns but also mitigates the potential risks of over-reliance on a single financial metric. These dual characteristics make GPA, as a profitability factor, more effective in market prediction than the HML factor. These features remain robust in the regression analysis of two sub-sample periods presented later.

In summary, the CAPM confirms the existence of the gross profitability anomaly, while the introduction of the RMW factor significantly enhances the performance of both the CH-5 and FF-5 models in explaining this anomaly. The CH-5 model performs better in the Chinese market and demonstrates a unique “crowding-out” effect.

To differentiate the impact of the shareholding reform, we adopt the same method as above, dividing the sample into two time periods: 2000.01-2007.06 and 2007.07-2024.11, and perform regression analysis, revealing different results. In [Table 8](#), the CAPM results exhibit similar characteristics to those discussed earlier, with the market factor showing stronger explanatory power in the pre-reform period, and the t-values in all three panels being significantly higher than those after the reform. The unexplained gross profitability excess returns are notably higher before the reform. At the same time, we observe that post-reform, investors pay more attention to the gross profitability anomaly, which has somewhat diminished its abnormal excess returns. This indicates that after the reform, investors’ investment thinking has become more mature, incorporating more pricing factors into their investment strategies.

The coefficients and t-values of the profitability factor show significant differences before and after the reform. Post-reform, the pricing efficiency of the RMW factor improves significantly. Specifically, in Panel B, the coefficient decreases from 1.61 to 1.25, suggesting a reduction in abnormal returns from gross profitability as the market matures. However, the t-value rises from 9.14 to 13.77, indicating that the market’s ability to price gross profitability anomaly returns improves even as investor attention to these anomalies has significantly increased. This feature remains robust across different weighting methods, with the SMB and CMA factors exhibiting similar characteristics, although with less pronounced changes compared to the RMW factor. These results suggest that the pricing efficiency of the RMW factor constructed using GPA improves during the mature market cycle, providing a more robust explanation for unexplained abnormal returns in the market, even as investors’ attention to these anomalies has noticeably increased.

7.2. Accrual anomaly

The accrual anomaly refers to stocks of companies with higher accruals, i.e., those with significant differences between accounting profits and cash flows, which tend to underperform in the future compared to companies with lower accruals. Studies have shown that companies often use accruals to manage earnings when reporting profits, which may obscure their actual financial condition. As a result, companies with high accruals typically have lower expected returns, while companies with low accruals tend to exhibit higher expected returns.

There are two primary approaches to measuring accruals: the balance sheet-based method and the cash flow statement-based method. Given the influence of financial statement quality in the Chinese market, and the fact that cash flow statements are more difficult to manipulate or falsify, this study adopts the cash flow statement method to calculate accruals. [Dechow \(1994\)](#) proposed a method for assessing earnings quality based on the cash flow statement, exploring the relationship between operating cash flow and accounting profits. His research showed that the differences between net income and cash flow can reveal the true quality of a company’s earnings, helping investors identify signs of earnings management. [Sloan \(1996\)](#) furthered this by examining the difference between operating cash flow and accruals, which reflects the quality of a company’s earnings. He argued that combining operating cash flow and accrual earnings can better predict future profitability and stock returns. Specifically, in the stock market, low-quality accruals (i.e., profits heavily reliant on accounting adjustments) are often associated with lower future returns.

Building on the work of these scholars, this paper uses adjusted accruals to measure the degree of earnings management in companies. The calculation method is as follows:

$$\text{Adjusted Accruals}_t = \frac{\text{Operating Profit}_t - \text{Operating Cash Flow}_t}{\text{Total Assets}_{t-1}} \quad (4)$$

The formula for Adjusted Accruals at time t is calculated by subtracting the operating cash flow from the operating profit for the current period t , and then dividing the result by the total assets at the end of the previous period $t - 1$. This measure aims to capture the accrual component of earnings that is not backed by cash flow, reflecting the extent to which a company’s earnings are driven by non-cash factors. By adjusting for operating cash flows, the formula provides a clearer picture of the company’s true profitability, excluding the effects of accruals that may distort the

Table 8

Alphas and factor loadings for gross profitability anomaly in different periods.

The table reports the monthly long-short return spread's alpha and factor loadings. The regression estimated is:

$$R_t = \alpha + \beta_{MKT} MKT_t + \beta_{SMB} SMB_t + \beta_{HML} HML_t + \beta_{RMW} RMW_t + \beta_{CMA} CMA_t + \epsilon_t$$

where R_t is the anomaly's long-short return spread in month t , MKT_t is the excess market return, SMB_t is size factor, and HML_t is the BM-based value factor, CMA_t is based on the investment factor using total assets, RMW_t is constructed using GPA. In Panel A, we use the market cap weighted method for sorting. In this approach, the long leg of the Gross profitability anomaly is the value-weighted portfolio of stocks in the highest decile of the anomaly measure, and the short leg contains the stocks in the lowest decile. A high value of the anomaly is associated with higher returns, so the long-short portfolios create a return spread. We use regression to examine the statistical significance of this return spread. In Panel B, we first remove the bottom 30% of the smallest stocks to avoid contamination from shell value and other small-cap effects. After this, we proceed with market cap weighting on the remaining sample. To distinguish the impact of the shareholding reform, we follow the method outlined above and divide the sample into two time periods. January 2000 to June 2007, and July 2007 to November 2024. The results show distinct differences between these periods. All t-statistics are based on the heteroskedasticity-consistent standard errors of [White \(1980\)](#).

Model	α	β_{MKT}	β_{SMB}	β_{HML}	β_{RMW}	β_{CMA}	$t(\alpha)$	$t(\beta_{MKT})$	$t(\beta_{SMB})$	$t(\beta_{HML})$	$t(\beta_{RMW})$	$t(\beta_{CMA})$
Panel A: Market Cap Weighted												
2000-2007												
CAPM	2.12	-0.28	—	—	—	—	2.69	-2.80	—	—	—	—
CH5	0.86	0.13	—	—	2.01	—	2.39	2.55	—	—	17.59	—
	0.80	0.13	-0.45	0.18	1.62	-0.19	2.22	2.50	-3.11	1.33	7.69	-0.68
2007-2024												
CAPM	0.99	-0.24	—	—	—	—	2.73	-4.95	—	—	—	—
CH5	0.41	-0.06	—	—	1.46	—	2.00	-2.13	—	—	20.26	—
	0.78	-0.07	-0.40	-0.10	1.24	-0.19	4.09	-2.55	-6.30	-1.25	14.29	-1.75
Panel B: Market Cap Weighted After Removing 30% Smallest Size												
2000-2007												
CAPM	1.72	-0.30	—	—	—	—	2.38	-3.22	—	—	—	—
CH5	0.53	0.09	—	—	1.89	—	1.86	2.26	—	—	20.83	—
	0.58	0.08	-0.29	0.03	1.61	-0.11	1.91	1.90	-2.38	0.22	9.14	-0.50
2007-2024												
CAPM	0.89	-0.26	—	—	—	—	2.45	-5.45	—	—	—	—
CH5	0.31	-0.08	—	—	1.47	—	1.51	-2.97	—	—	20.30	—
	0.67	-0.09	-0.37	-0.13	1.25	-0.14	3.32	-3.36	-5.56	-1.57	13.77	-1.22

financial performance in the short term. The use of the prior period's total assets serves to normalize the measure, allowing for a consistent comparison across firms of different sizes.

[Table 9](#) presents the results of our test for the existence of the earnings management anomaly and its performance in the Chinese market, using long-short portfolios where the group with low earnings management serves as the long position, and the group with high earnings management serves as the short position. Unlike the previous results for the gross profitability anomaly, the CH-5 model performs well in explaining the earnings management anomaly when using only the market factor (MKT) and the profitability factor as explanatory variables. According to Panel B of the

table, the alpha coefficient of the CH-5 model is 0.30, with a t-value of 1.14, indicating that under certain conditions, the model can significantly capture abnormal returns related to earnings management in the market after excluding the influence of other factors. This suggests that the CH-5 model, using only the market factor and the profitability factor, can partially reveal the existence of earnings management in the market.

Table 9

Alphas and factor loadings for accrual anomaly.

The table reports the monthly long-short return spread's alpha and factor loadings. The regression estimated is:

$$R_t = \alpha + \beta_{MKT} MKT_t + \beta_{SMB} SMB_t + \beta_{HML} HML_t + \beta_{RMW} RMW_t + \beta_{CMA} CMA_t + \epsilon_t$$

where R_t is the anomaly's long-short return spread in month t , MKT_t is the excess market return, SMB_t is size factor, and HML_t is the BM-based value factor, CMA_t is based on the investment factor using total assets, RMW_t is constructed using both ROE and GPA, but in this table, RMW_t is used to represent both. In Panel A, we use the market cap weighted method for sorting. In this approach, the long leg of the accrual anomaly is the value-weighted portfolio of stocks in the lowest decile of the anomaly measure, and the short leg contains the stocks in the highest decile. A high value of the anomaly is associated with lower returns, so the long-short portfolios create a return spread. We use regression to examine the statistical significance of this return spread. In Panel B, we first remove the bottom 30% of the smallest stocks to avoid contamination from shell value and other small-cap effects. After this, we proceed with market cap weighting on the remaining sample. Our sample period is January 2000 through November 2024. All t-statistics are based on the heteroskedasticity-consistent standard errors of [White \(1980\)](#).

Model	α	β_{MKT}	β_{SMB}	β_{HML}	β_{RMW}	β_{CMA}	$t(\alpha)$	$t(\beta_{MKT})$	$t(\beta_{SMB})$	$t(\beta_{HML})$	$t(\beta_{RMW})$	$t(\beta_{CMA})$
Panel A: Market Cap Weighted												
CAPM	0.63	-0.05	—	—	—	—	2.13	-1.33	—	—	—	—
CH5	0.27	0.08	—	—	0.87	—	1.05	2.15	—	—	9.82	—
	0.58	-0.02	0.22	-0.73	0.47	-0.4	3.01	-0.89	3.42	-9.40	5.02	-0.08
FF5	0.54	-0.07	—	—	0.30	—	1.87	-0.17	—	—	3.04	—
	0.66	-0.03	0.30	-0.91	0.51	0.19	3.49	-0.99	4.07	-13.36	5.21	1.49
Panel B: Market Cap Weighted After Removing 30% Smallest Size												
CAPM	0.66	-0.07	—	—	—	—	2.21	-1.69	—	—	—	—
CH5	0.30	0.06	—	—	0.88	—	1.14	1.76	—	—	9.80	—
	0.61	-0.03	0.21	-0.72	0.48	-0.4	2.98	-1.21	2.98	-8.72	5.02	-0.04
FF5	0.57	-0.02	—	—	0.32	—	1.94	-0.46	—	—	3.15	—
	0.70	-0.04	0.27	-0.91	0.51	0.18	3.49	-1.39	3.52	-12.48	4.87	1.37

However, when the HML factor is introduced, the model's explanatory power changes significantly. According to the table, after including the HML factor, the alpha coefficient of the CH-5 model increases to 0.61, and the t-value rises to 2.98 (Panel B). This change suggests that the inclusion of the HML factor makes the alpha value more positive, even restoring it to a state where it is no longer related to earnings management. This means that the HML factor strongly interferes with the earnings management effect in the market, possibly because of the "crowding-out effect" of the RMW factor, which causes the HML factor to mistakenly capture certain characteristics in value stocks, characteristics that are somewhat confounded with the earnings management phenomenon. This, in turn, affects the model's ability to explain earnings management and causes the alpha value in the regression to lose significance. This phenomenon contrasts with the explanation process of the gross profitability anomaly. In the case of the gross profitability anomaly, the RMW factor, as a proxy for profitability, effectively explains the related excess returns and

remains significant in both market-cap-weighted and small-cap-excluded portfolios. These results are also robust across different classification methods.

Furthermore, we observe the differential impact of market capitalization screening on the earnings management anomaly. The results for equal-weighted and market-cap-weighted excess returns are reported in the Appendix. The alpha coefficient for the CAPM model is 0.24, with a t-value of 1.16, suggesting that, without market cap screening, the earnings management anomaly is weak in small samples. Although the alpha coefficient is positive, the t-value is close to 1, indicating low significance, and thus the result fails to effectively capture excess returns associated with earnings management. However, when using the market-cap-weighting method, the alpha coefficient and t-value for the CAPM model increase in both Panel A and Panel B. For example, in Panel A, the alpha coefficient for the CAPM model is 0.63, with a t-value of 2.13, indicating that in large-cap stocks, the impact of earnings management is more significant and can significantly reflect the abnormal returns associated with earnings management. In Panel B, after excluding the bottom 30% of small-cap stocks, the alpha coefficient further increases to 0.66, with a t-value of 2.21, continuing to show the significance of the earnings management anomaly in large-cap stocks.

These results suggest that the market cap screening method has a significant effect on the earnings management anomaly. Specifically, after applying market cap weighting and excluding small-cap stocks, the excess returns from earnings management anomalies are better explained in large-cap companies. This phenomenon may arise because small-cap companies engage in less earnings management, and their abnormal returns are more easily identified by the market. Therefore, market cap weighting helps to better reveal the earnings management effects in large-cap stocks, further confirming that in mature markets, large-cap companies typically exhibit stronger earnings management abilities, and their abnormal returns are harder to identify.

In terms of model comparison, the FF-5 model performs marginally compared to the CH-5 model. From the table, we can see that although the alpha coefficients for the FF-5 model are positive in all panels, their t-values are generally lower, indicating relatively weak explanatory power, especially in comparison to the CH-5 model. In Panels A and B, the alpha coefficients and t-values for the FF-5 model do not show significant improvements compared to CAPM, suggesting that adding other factors did not provide strong support for explaining the earnings management anomaly. By contrast, the CH-5 model performs better in capturing the earnings management anomaly, particularly when no additional factors are included. In comparison, the FF-5 model shows limited performance, especially in explaining the earnings management anomaly, and does not exhibit the significant explanatory power demonstrated by the CH-5 model. We also perform subsample regression analysis for different periods, with results similar to those discussed above. The detailed results of the Accrual anomaly regressions for subsamples are presented in the Appendix.

8. Conclusion

Due to the relatively late development of China's capital market, the quality of financial reporting by Chinese listed companies has been severely affected by issues such as earnings management and policy interventions, which significantly distort the pricing effectiveness of profitability factors. This problem has become even more pronounced following a prolonged period under the split-share structure system. Against this backdrop, this study conducts a "horse race" among various profitability factors and finds that the GPA factor demonstrates superior explanatory power for profitability effects compared to the traditional ROE factor. Subsequently, using Fama-French's 2×3 sorting method, this study constructs the CH-5 factor model, replacing ROE with GPA as the profitability proxy. Through a series of regression analyses, the results show that the CH-5 model based on GPA more accurately reflects firms' core profitability and filters out distortions caused by earnings management and policy interventions, providing more robust support for asset pricing.

Empirical results further demonstrate that the CH-5 model outperforms the traditional Fama-French five-factor model in explaining excess returns. Notably, the inclusion of GPA significantly enhances the model's ability to capture market return fluctuations, thereby improving its pricing efficiency.

In addition, this study examines factor redundancy tests and finds that the explanatory power of the CMA factor and other factors has shifted significantly before and after China's split-share structure reform. Specifically, the size and profitability effects have become noticeably stronger in the post-reform period. Following this logic, we also perform sub-sample tests across different time periods in subsequent analyses, which yield varying results. These findings further confirm the necessity of localized adjustments to asset pricing models in the Chinese market. This structural shift is not only relevant to the context of this study but may also provide valuable insights for future factor pricing research, particularly regarding the temporal variations caused by China's split-share reform.

Furthermore, through regression analysis on two well-documented market anomalies in China—the gross profitability anomaly and the accrual anomaly—we find that the CH-5 model, which incorporates GPA, can significantly explain both anomalies, whereas the FF-5 model fails to do so effectively. In particular, regarding the gross profitability anomaly, GPA, despite being a profitability factor, exhibits a dual role by capturing both profitability and value effects. This allows GPA to not only reflect firm profitability but also mitigate the traditional value strategy’s issue of selecting low-quality assets, thereby enhancing the model’s ability to predict returns related to firm valuation. While this study highlights this dual effect, it does not fully explore the potential “crowding-out effect” of the profitability factor, which warrants further investigation, especially under different policy regimes.

Overall, this research provides a novel perspective for improving the Fama-French five-factor model and offers a localized asset pricing framework for the Chinese market. Investors may also consider using GPA as an alternative profitability factor to optimize portfolio construction. Future research could explore the applicability of the GPA factor in other emerging markets. As global economic dynamics evolve, emerging markets such as those in Southeast Asia share similarities with the early stages of China’s capital market, raising the question of whether similar pricing characteristics may also emerge in these markets—an avenue worthy of further exploration.

A. Appendix

A.1. Fama-MacBeth regressions under different samples

To further validate the robustness of the regression analysis results, this study conducted robustness tests on the Fama-MacBeth regression. Table 10 and Table 11 present the results of the monthly Fama-MacBeth regression analysis for the Shanghai Stock Exchange and the Shanghai and Shenzhen Stock Exchanges, respectively. The regression factors include market beta, logME, book-to-market ratio, profitability factors ROE and GPA, and investment factor INV. The average slopes and t-values for each valuation factor when added individually to the model are listed, aiming to test the explanatory power of each factor in the Chinese market.

The test results indicate that the GPA factor significantly outperforms the ROE factor in the regression analysis. The GPA factor continues to exhibit significant explanatory power and consistently dominates in all regression models, while the performance of ROE remains insignificant. These robustness test results suggest that the GPA factor has higher stability and predictive ability in explaining stock returns in the Chinese market. Compared to ROE, GPA effectively removes the interference of factors such as earnings management, providing more robust support for asset pricing models.

A.2. Cross-Explanatory Power Between CH-3 and CH-5 Factors

We further compare the explanatory power of the CH-5 and CH-3 models to each other, with the results reported in Table 12.

Panel A presents the estimated monthly alphas of one model’s factors with respect to the other model. The results show that when the CH-3 factors (SMB and VMG) are tested against the CH-5 model, the alphas are economically small and statistically insignificant (t-values of -0.86 and 1.08 , respectively). This indicates that the CH-5 model can adequately explain the CH-3 factors. In contrast, when the CH-5 factors are tested against the CH-3 model, the alphas for SMB and RMW are significantly positive (t-values of 2.60 and 2.69 , respectively), while CMA is positive but statistically insignificant. These results suggest that the CH-3 model cannot fully capture several of the CH-5 factors.

Panel B reports the results of the Gibbons–Ross–Shanken (1989) test. When assessing the ability of the CH-5 model to explain the CH-3 factors, the GRS F-statistic is 1.74 with a p-value of 0.18 , indicating that we cannot reject the null hypothesis that all alphas are jointly zero. This suggests that the CH-5 model adequately captures the CH-3 factors. By contrast, when examining the CH-3 model’s ability to explain the CH-5 factors, the GRS statistic is 5.55 with a p-value of 0.0002 , strongly rejecting the null hypothesis and indicating that the CH-3 model performs poorly in explaining the CH-5 factors.

Overall, these findings demonstrate that the CH-5 model subsumes the CH-3 model, while the reverse does not hold. This highlights the incremental explanatory power of the CH-5 model in asset pricing, particularly in capturing profitability and investment dimensions.

A.3. Anomaly tests under different weighting methods

Table 13 presents the results of the Gross Profitability Anomaly test using the equal-weighted method over different periods and all periods combined, employing the CH-5 model for analysis. The results indicate that the CH-5 model

performs exceptionally well in explaining the excess returns that the CAPM model fails to capture, which is consistent with earlier findings. The table summarizes the alpha and factor loadings for the gross profitability anomaly during three distinct periods: 2000-2007, 2007-2024, and the full sample from January 2000 to November 2024.

In contrast, the CH-5 model demonstrates stronger explanatory power, with alpha values reduced across all periods, and the RMW factor exhibiting higher t-values when explaining the abnormal returns. These results indicate that the CH-5 model, particularly the profitability factor, effectively accounts for the abnormal returns associated with the gross profitability anomaly.

Table 13 presents the results of the accrual anomaly tests under three different weighting methods. The CAPM alpha is higher among large-cap firms compared to small-cap firms, suggesting that the accrual anomaly yields greater long-short excess returns in the large-cap sample. Additionally, the anomaly exhibits distinct patterns across different sample periods, with alpha values significantly higher in the post-reform period compared to the pre-reform period, consistent with the findings discussed earlier.

The RMW factor shows a relatively high t-value in explaining this anomaly; however, its explanatory power diminishes when additional factors are included in the model, which is consistent with the “crowding-out effect” discussed previously.

Table 10

Fama-MacBeth regressions of stock returns on valuation factors.

The table reports average slope coefficients from month-by-month Fama-MacBeth regressions. Individual stock returns are regressed cross-sectionally on stock characteristics as of the previous month. The first to fifth columns present the regression results based on the factors used in the Fama-French five-factor (FF-5) model. The sixth and seventh columns report the regressions employing the CH-5 factors. The eighth column displays the results of a horse race regression that incorporates all factors together. The columns correspond to different regression specifications, with nonempty rows indicating the included regressors. The sample period is January 2000 through December 2024. Our sample data includes only stocks codes starting with 600 and 000.

Quantity	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Intercept</i>	0.0026 (1.20)	-0.0708 (-4.66)	-0.0674 (-4.42)	-0.0677 (-4.45)	-0.0695 (-4.64)	-0.0650 (-4.33)	-0.0669 (-4.51)	-0.0671 (-4.53)
<i>Beta</i>	0.0049 (1.99)	-0.0415 (-3.33)	-0.0394 (-3.20)	-0.0396 (-3.23)	-0.0413 (-3.43)	-0.0382 (-3.20)	-0.0400 (-3.40)	-0.0401 (-3.42)
<i>logME</i>	- (4.86)	0.0064 (4.73)	0.0062 (4.77)	0.0062 (5.03)	0.0064 (4.63)	0.0059 (4.89)	0.0061 (4.90)	0.0061
<i>BM</i>	- (-3.84)	- (-3.84)	-0.0045 (-3.84)	-0.0045 (-3.84)	-0.0043 (-3.65)	-0.0043 (-3.80)	-0.0041 (-3.57)	-0.0041 (-3.56)
<i>ROE</i>	- (-0.79)	- (-0.79)	- (-0.79)	-0.0003 (-0.79)	-0.0002 (-0.42)	- (-0.42)	- (-0.42)	-0.0002 (-0.53)
<i>GPA</i>	- (3.36)	- (3.36)	- (3.36)	- (3.36)	- (3.36)	0.0557 (3.36)	0.0571 (3.48)	0.0571 (3.49)
<i>INV</i>	- (-2.53)	- (-2.53)	- (-2.53)	- (-2.53)	-0.0040 (-2.53)	- (-2.53)	-0.0043 (-2.82)	-0.0043 (-2.82)

Table 11

Fama-MacBeth regressions of stock returns on valuation factors.

The table reports average slope coefficients from month-by-month Fama-MacBeth regressions. Individual stock returns are regressed cross-sectionally on stock characteristics as of the previous month. The first to fifth columns present the regression results based on the factors used in the Fama-French five-factor (FF-5) model. The sixth and seventh columns report the regressions employing the CH-5 factors. The eighth column displays the results of a horse race regression that incorporates all factors together. The columns correspond to different regression specifications, with nonempty rows indicating the included regressors. The sample period is January 2000 through December 2024. Our sample data includes only stocks codes starting with 600.

Quantity	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Intercept</i>	0.0024 (1.12)	-0.0759 (-4.98)	-0.0717 (-4.69)	-0.0720 (-4.72)	-0.0736 (-4.90)	-0.0693 (-4.61)	-0.0710 (-4.79)	-0.0583 (-4.07)
<i>Beta</i>	0.0048 (1.92)	-0.0370 (-2.89)	-0.0348 (-2.74)	-0.0349 (-2.76)	-0.0366 (-2.94)	-0.0335 (-2.71)	-0.0352 (-2.90)	-0.0387 (-3.39)
<i>logME</i>	-	0.0067 (4.93)	0.0063 (4.72)	0.0063 (4.75)	0.0065 (5.00)	0.0060 (4.60)	0.0062 (4.83)	0.0055 (4.61)
<i>BM</i>	-	-	-0.0051 (-3.93)	-0.0051 (-3.93)	-0.0050 (-3.76)	-0.0048 (-3.86)	-0.0047 (-3.67)	-0.0037 (-3.80)
<i>ROE</i>	-	-	-	-0.0002 (-0.60)	-0.0001 (-0.25)	-	-	-0.0004 (-0.74)
<i>GPA</i>	-	-	-	-	-	0.0613 (3.57)	0.0624 (3.66)	0.0487 (2.95)
<i>INV</i>	-	-	-	-	-0.0033 (-2.16)	-	-0.0371 (-2.46)	-0.0025 (-1.56)

Table 12

Abilities of models CH-5 and CH-3 to explain each other's factors.

Panel A reports a factor's estimated monthly alpha with respect to the other model (with [White \(1980\)](#) heteroskedasticity-consistent t-statistics in parentheses). Panel B reports the Gibbons–Ross–Shanken ([Gibbons et al., 1989](#)) F-test of whether a given model produces zero alphas for the factors of the other model (p-value in parentheses). The GRS statistic assesses whether a regression model significantly deviates from zero alpha as a whole. A higher F-value indicates that the model has weaker explanatory power for the factors of another model, meaning it cannot fully account for the factor returns of the other model. Conversely, a higher p-value suggests that the model can adequately explain the factor returns of the other model. The sample period is January 2000 through December 2023. For the construction of the CH-3 model, see [Liu et al. \(2019\)](#).

Panel A: Alpha (t-statistic)		
Alphas with respect to:	CH-5	CH-3
CH3-SMB	-0.0157 (-0.86)	– –
CH3-VMG	0.1038 (1.08)	– –
SMB	– –	0.0435 (2.60)
RMW	– –	0.4236 (2.69)
Panel B: GRS F-statistics (p-value)		
Alphas with respect to:	CH-5	CH-3
CH-3(SMB,VMG)	1.74 (0.18)	– –
SMB,HML,RMW,CMA	– –	5.55 (0.0002)

Table 13

Alphas and factor loadings for gross profitability anomaly in equal weighted.

The table reports the monthly long-short return spread's alpha and factor loadings. The regression estimated is:

$$R_t = \alpha + \beta_{MKT} MKT_t + \beta_{SMB} SMB_t + \beta_{HML} HML_t + \beta_{RMW} RMW_t + \beta_{CMA} CMA_t + \epsilon_t$$

We use the equal weighted method for sorting. In this approach, the long leg of the Gross profitability anomaly is the value-weighted portfolio of stocks in the highest decile of the anomaly measure, and the short leg contains the stocks in the lowest decile. A high value of the anomaly is associated with higher returns, so the long-short portfolios create a return spread. We use regression to examine the statistical significance of this return spread. Our sample period is January 2000 through November 2024. All t-statistics are based on the heteroskedasticity-consistent standard errors of [White \(1980\)](#).

Model	α	β_{MKT}	β_{SMB}	β_{HML}	β_{RMW}	β_{CMA}	$t(\alpha)$	$t(\beta_{MKT})$	$t(\beta_{SMB})$	$t(\beta_{HML})$	$t(\beta_{RMW})$	$t(\beta_{CMA})$
CAPM												
All periods	1.61	-0.19	—	—	—	—	5.15	-4.59	—	—	—	—
2000-2007	2.18	-0.26	—	—	—	—	2.80	-2.59	—	—	—	—
2007-2024	1.42	-0.16	—	—	—	—	4.55	-3.96	—	—	—	—
CH5												
All periods	0.99	0.02	—	—	1.42	—	5.21	0.65	—	—	22.11	—
	1.15	0.01	-0.19	-0.07	1.26	-0.07	5.76	0.35	-2.91	-0.97	13.56	-0.61
2000-2007	1.00	0.13	—	—	1.88	—	2.37	2.16	—	—	14.08	—
	0.85	0.14	-0.34	0.24	1.60	-0.21	1.93	2.24	-1.92	1.48	6.24	-0.63
2007-2024	0.94	-0.02	—	—	1.21	—	4.89	-0.66	—	—	17.90	—
	1.14	-0.03	-0.16	-0.19	1.05	0.04	5.57	-1.07	-2.33	-2.28	11.27	0.33
FF5												
All periods	1.28	-0.02	—	—	1.13	—	5.37	-0.72	—	—	14.19	—
	1.60	-0.05	-0.20	-0.55	0.95	0.12	7.07	-1.64	-2.36	-7.04	8.09	0.82

Table 14

Alphas and factor loadings for accrual anomaly.

The table reports the monthly long-short return spread's alpha and factor loadings. The regression estimated is:

$$R_t = \alpha + \beta_{MKT} MKT_t + \beta_{SMB} SMB_t + \beta_{HML} HML_t + \beta_{RMW} RMW_t + \beta_{CMA} CMA_t + \epsilon_t$$

In this approach, the long leg of the accrual anomaly is the value-weighted portfolio of stocks in the lowest decile of the anomaly measure, and the short leg contains the stocks in the highest decile. A high value of the anomaly is associated with lower returns, so the long-short portfolios create a return spread. We use regression to examine the statistical significance of this return spread. In panel A, we use the equal weighted method for sorting. In Panel B, we use the market cap weighted method for sorting. In Panel C, we remove the bottom 30% of the smallest stocks to avoid contamination from shell value and other small-cap effects. Our sample period is January 2000 through November 2024. All t-statistics are based on the heteroskedasticity-consistent standard errors of [White \(1980\)](#).

Model	α	β_{MKT}	β_{SMB}	β_{HML}	β_{RMW}	β_{CMA}	$t(\alpha)$	$t(\beta_{MKT})$	$t(\beta_{SMB})$	$t(\beta_{HML})$	$t(\beta_{RMW})$	$t(\beta_{CMA})$
Panel A: Equal Weighted												
CAPM												
2000-2007	0.24	-0.13	–	–	–	–	0.79	-3.41	–	–	–	–
2007-2024	0.74	-0.08	–	–	–	–	2.56	-2.11	–	–	–	–
CH-5												
2000-2007	-0.03	-0.02	–	–	0.59	–	-0.11	-0.62	–	–	7.31	–
	0.14	-0.04	0.16	-0.26	0.54	-0.15	0.57	-1.08	1.49	-2.88	3.61	-0.83
2007-2024	0.40	0.02	–	–	0.86	–	1.72	0.71	–	–	10.54	–
	0.40	-0.04	0.23	-0.32	0.64	0.00	1.96	-1.28	3.26	-3.80	6.83	0.08
Panel B: Market Cap Weighted												
CAPM												
2000-2007	0.33	-0.08	–	–	–	–	0.84	-1.65	–	–	–	–
2007-2024	0.79	-0.08	–	–	–	–	2.15	-1.60	–	–	–	–
CH-5												
2000-2007	0.15	-0.08	–	–	0.54	–	0.40	-0.16	–	–	4.19	–
	0.45	-0.02	0.27	-0.58	0.54	0.01	1.39	-0.53	1.96	-4.85	2.65	0.05
2007-2024	0.38	0.05	–	–	0.51	–	1.25	1.10	–	–	9.64	–
	0.56	-0.07	0.25	-0.58	0.51	-0.23	2.32	-2.25	3.10	-5.85	4.70	-1.70
Panel C: Market Cap Weighted After Removing 30% Smallest Size												
CAPM												
2000-2007	0.39	-0.11	–	–	–	–	0.95	-2.06	–	–	–	–
2007-2024	0.71	-0.09	–	–	–	–	1.90	-1.82	–	–	–	–
CH-5												
2000-2007	0.09	0.02	–	–	0.54	–	0.25	0.35	–	–	4.48	–
	0.52	-0.05	0.28	-0.58	0.51	-0.05	1.46	-0.98	1.84	-4.43	2.32	-0.19
2007-2024	0.30	0.05	–	–	0.88	–	1.14	1.76	–	–	9.80	–
	0.50	-0.09	0.21	-0.56	0.49	-0.27	1.95	-2.44	2.47	-5.30	4.22	-1.87

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